



Vivekanand Education Society's Institute of Technology

(Affiliated to University of Mumbai, Approved by AICTE & Recognized by Govt. of Maharashtra)

CalorEase

Submitted in partial fulfilment of the requirements

of the

T.E Project in
Machine Learning

by

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under the guidance of

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Department of Artificial Intelligence and Data Science
Vivekanand Education Society's Institute of Technology
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Department of Artificial Intelligence and Data Science

CERTIFICATE

This is to certify that **Mr. Soham Kapadi & Mr. Varun Shukla** of T.E D11AD Div B of Artificial Intelligence and Data Science studying under the University of Mumbai have satisfactorily presented the Project entitled **CalorEase** as a part of the T.E Mini Project for Semester-VI of Machine Learning Lab under the guidance of **Bincy Ivin** in the year 2024-2025.

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DECLARATION

We, *Soham Kapadi, Varun Shukla* from *class D11AD-B*, declare that this project represents our ideas in our own words without plagiarism and wherever others' ideas or words have been included, we have adequately cited and referenced the original sources.

We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our project work.

We declare that we have maintained a minimum 75% attendance, as per the University of Mumbai norms.

Yours Faithfully

1. _____

2. _____

3. _____

(Name & Signature of Students with Date)

Acknowledgement

We extend my deepest gratitude to our esteemed faculty members, especially **Mrs. Sangeeta Oswaal** and **Ms. Bincy Ivin**, for their unwavering guidance, insightful feedback, and continuous support throughout this project. Their expertise and encouragement have been invaluable in shaping this work.

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ABSTRACT

Managing calorie balance effectively is a crucial aspect of maintaining a healthy lifestyle. Many individuals struggle with determining how long they need to work out to burn off the calories they consume, leading to inefficient or unsustainable exercise routines. *Calorease* is a fitness-focused system designed to address this issue by providing personalized workout duration estimates based on key user metrics, including height, weight, and calorie intake. By leveraging machine learning algorithms such as Linear Regression, Random Forest, and Gradient Boosting, the system predicts the optimal duration required for an individual to burn a specific number of calories through exercise.

The project is built on a structured dataset containing various exercises and their respective calorie-burning rates. Exploratory Data Analysis (EDA) was conducted to understand the correlations between body parameters, calorie intake, and workout duration. The machine learning models were evaluated using performance metrics like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to ensure accurate and reliable predictions. Additionally, a recommendation engine has been integrated into *Calorease* to suggest the most suitable exercises for users based on their fitness goals and body composition.

To enhance accessibility, *Calorease* is deployed as an interactive Streamlit web application, allowing users to enter their details and receive real-time workout recommendations. Future improvements to the system include expanding the dataset with a broader range of exercises, refining calorie-burning calculations, and integrating real-time tracking through wearable fitness devices. By providing data-driven and personalized workout recommendations, *Calorease* empowers users to optimize their fitness routines, maintain a healthy calorie balance, and make informed decisions about their exercise habits.

1. Introduction

1.1 Overview of the project

In today's fast-paced world, staying fit and maintaining a healthy lifestyle can be challenging. *Calorease* is designed to simplify this process by providing a personalized approach to calorie burning. The system takes in a person's height, weight, and calorie intake to determine how much time they need to work out to burn those calories effectively. Whether someone is looking to lose weight, maintain their fitness, or simply be more conscious of their energy expenditure, *Calorease* helps by offering clear and actionable workout recommendations.

1.2 Scope of the project

The project focuses on delivering a user-friendly and efficient way to track calorie burn based on individual body metrics. It aims to assist people of all fitness levels by:

1. Calculating the estimated time required to burn a specific amount of calories.
2. Providing tailored workout recommendations based on height and weight.
3. Helping users make informed decisions about their exercise routines.

2. Objective & Problem Statement

2.1 Clear definition of the problem statement

Many people struggle to determine how much exercise is needed to burn off the calories they consume daily. Without proper guidance, they may either overwork themselves or not exercise enough, leading to ineffective weight management. Additionally, existing calorie calculators often provide generic estimates without considering individual body metrics like height and weight. This lack of personalization makes it difficult for users to get accurate workout duration recommendations that align with their fitness goals.

2.2 Objective of the project

The primary objective of *Calorease* is to provide a personalized and accurate estimate of workout duration needed to burn calories based on an individual's height, weight, and calorie intake. Specifically, the project aims to:

1. Offer a user-friendly system that calculates workout duration efficiently.
2. Provide personalized recommendations that cater to different body types.
3. Help users make informed decisions about their exercise routine for better health and fitness management.
4. Encourage consistency in fitness by making calorie tracking easier and more accessible.

By achieving these objectives, *Calorease* will help users maintain a balanced lifestyle and stay on track with their fitness goals.

3. Dataset Description & Collection

The dataset used for the Calorease project plays a foundational role in accurately estimating the duration of workouts required to burn a specified number of calories based on individual user metrics such as height, weight, and daily calorie intake. At its core, the dataset includes a diverse set of attributes that are closely related to physical activity and energy expenditure. These attributes encompass various types of exercises, their respective calorie-burning rates, and how different body parameters influence the overall energy consumption during workouts. The original data was sourced from Kaggle, a widely trusted platform known for providing rich and reliable datasets for data science and machine learning projects. Kaggle's datasets are often curated with contributions from fitness professionals and researchers, making them an ideal starting point for health and fitness-related applications. However, in its raw form, the dataset had limitations in terms of completeness, coverage, and adaptability for personalized recommendations.

To overcome these limitations and better suit the needs of Calorease, the dataset underwent an extensive data cleaning and enhancement process. This involved removing any inconsistencies, handling missing or duplicate entries, and standardizing the data format to ensure seamless integration into the machine learning pipeline. In addition to cleaning, the dataset was expanded while keeping the original structure and pattern intact. This expansion involved the inclusion of more exercise categories, a wider range of intensity levels, and updated calorie-burning metrics based on the latest findings from fitness literature. By augmenting the dataset with more diverse and detailed entries, we enabled the model to account for various user profiles and workout preferences, thereby increasing the accuracy and relevance of the predictions. The resulting dataset not only retains the reliability of the original Kaggle source but also offers enhanced versatility and depth, making it a powerful resource for developing a personalized calorie-burning estimation tool like Calorease.

4. Literature Survey

4.1 Literature/Techniques studied.

Predicting calories burned based on exercise time involves statistical and machine learning techniques. Traditional models like linear regression estimate calorie expenditure using exercise duration and intensity, while advanced approaches such as random forest and gradient boosting enhance accuracy. Wearable sensors provide real-time physiological data, improving predictions. Feature engineering, including heart rate and metabolic equivalent (MET), refines models. Hybrid techniques combining machine learning and physiological equations further enhance reliability, making calorie prediction more precise for fitness tracking and health monitoring.

4.2 Papers/Findings (Minimum 10 research papers to be referred)

1.A non invasive method for calculating calories burned during exercise using heartbeat

Author - G.Karthik Reddy, K. L. Achari

Key Findings -

Male: $((-55.0969 + (0.6309 \times \text{HR}) + (0.1988 \times \text{W}) + (0.2017 \times \text{A})) 4.184) \times 60 \times \text{T}$

Female: $((-20.4022 + (0.4472 \times \text{HR}) - (0.1263 \times \text{W}) + (0.074 \times \text{A})) 4.184) \times 60 \times \text{T}$

2. Bloom Balance: Calorie Balancing Application With Scientific Validation

Authors - Benjarat Tirasirichai; Peeraya Thanomboon; Pimpaknat Soontorntham;
Worapan Kusakunniran; Mark Robinson

Key Findings - $\text{Kcal/min} = [\text{VO}_2 \times 3.5 \times (\text{bodyweight in kg} / 1000)] \times 5$

3. Estimation of Calories Consumption for Aerobics Using Kinect Based Skeleton Tracking

Authors - Pei-Fu Tsou; Chao-Cheng Wu

Key Findings -

Formula	Main parameter	Accuracy
(Male) Calorie = $((-95.7735 + (0.634 \times \text{HR}) + (0.404 \times \text{VO}_{2\text{max}}) + (0.394 \times \text{W}) + (0.271 \times \text{A}))/4.184) \times 60 \times \text{T}$	$\text{VO}_{2\text{max}}$ & HR	High
(Female) Calorie = $((-59.3954 + (0.45 \times \text{HR}) + (0.380 \times \text{VO}_{2\text{max}}) + (0.103 \times \text{W}) + (0.274 \times \text{A}))/4.184) \times 60 \times \text{T}$	$\text{VO}_{2\text{max}}$ & HR	High
(Male) Calorie = $((-55.0969 + (0.6309 \times \text{HR}) + (0.1988 \times \text{W}) + (0.2017 \times \text{A}))/4.184) \times 60 \times \text{T}$	HR	Middle
(Female) Calorie = $((-20.4022 + (0.4472 \times \text{HR}) - (0.1263 \times \text{W}) + (0.074 \times \text{A}))/4.184) \times 60 \times \text{T}$	HR	Middle
Calorie = $\text{MET} \times \text{W} \times \text{T}$	MET	Low

Table 4.2.1(Estimation of Calories)

4. Very low-frequency heart rate variability wave amplitude and sympathetic stimulation-characterization and modeling

Authors - R.J. Leor-Librach; S. Eliash; E. Kaplinsky; B.-Z. Bobrovsky

Key Findings - The purpose of this study was to assess the relationship between very low-frequency heart rate variability (LFHR) wave amplitude and the degree of sympathetic stimulation. We developed a computerized system for the controlled increase of heart rate (HR) by isoproterenol (ISP), with which we obtained a series of stabilized HR levels in conscious freely moving rats. We found that LFHR amplitude rises gradually as a function of the average HR for each level until it reaches a point where additional increases in average HR are associated with gradual decrease in LFHR amplitude. We successfully built and fitted a model of LFHR amplitude to the experimental results. The fact that our model fits the experimental data well may suggest

a possible relationship between our LFHR amplitude findings and the basic physiologic properties of the HR-ISP system inherent in our model.

5. Very low-frequency heart rate variability wave amplitude and sympathetic stimulation-characterization and modeling

Authors - G Naga Venkata RamaKrishna; S Sreelakshmi; Meenakshi Diwakar; S. Ramkumar; T Vinithra Banu; T. Thilagam

Key Findings - The technique that has been recommended, which is known as the Improved Learning Model for Calorie Prediction (ILMCP), has the most accurate findings in terms of mean absolute error. The ILMCP approach is utilized by the system in order to segment the picture in order to label the object. This is accomplished by employing a box. As a means of ensuring that the object is distinguished from the background, the box is employed. When viewed from the outside, the sides of the rectangle represent the backdrop, while the interior of the algorithm is a combination of the object and other backdrops. The following stage is to use an iterative strategy that identifies each pixel as either the background or the foreground. This will be the next approach. The computer does the first tagging based on the information provided by the user. After that, we use the Gaussian mixture model to reconstruct the background and the foreground of the environment.

6. Implementing Calorie Burnt Prediction Through XGBOOST Algorithm Compared Over Random Forest Regressor

Authors - K Hima Vamsi Reddy; Terrance Frederick Fernandez

Key Findings -

(SPSS) to handle data when it's suitable for statistical analysis. As an example, you may evaluate XGBoost and Novel Random Forest Regressor's ability to estimate energy usage in a specific situation using SPSS. Using SPSS version 21, we will apply the outcomes of the first two sets of algorithms. After comparing XGBoost with Novel Random Forest Regressor, we used a t-test for independent samples.

7. Implementing Calorie Burnt Prediction Through XGBOOST Algorithm Compared Over Random Forest Regressor

Authors - K Hima Vamsi Reddy; Terrance Frederick Fernandez

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(SPSS) to handle data when it's suitable for statistical analysis. As an example, you may evaluate XGBoost and Novel Random Forest Regressor's ability to estimate energy usage in a specific situation using SPSS. Using SPSS version 21, we will apply the outcomes of the first two sets of algorithms. After comparing XGBoost with Novel Random Forest Regressor, we used a t-test for independent samples.

8. Burned Calories Prediction using Supervised Machine Learning: Regression Algorithm

Authors - Marte Nipas; Aimee G. Acoba; Jennalyn N. Mindoro; Mon Arjay F. Malbog; Julie Ann B. Susa; Joshua S. Gulmatico

Key Findings -

Regression models are one of the Machine Learning algorithms used to make a prediction. In this study it was able to predict the calories burned by an individual using a given dependent variable. After the identification of the problem, cleaning and preparation the data are needed before it can be fed on the algorithm. This study needs to be analyzed to determine the relationship present between each variable. It is important to visualize and understand the relationships between variables to look for problems like multicollinearity and other issues. Because those problems have an impact on the algorithm that will be used. Plotting the samples in a 2-d plane can give us insights. In this study the researchers saw that the target variable has a visible correlation between heart rate, exercise duration, and body temperature. Lastly, the training and selection model is important and must be carefully conducted. Choosing the best fit model will make the prediction more accurate and reliable. The researchers achieve 95.77% accuracy using the random forest regression algorithm with the following hyper parameters: estimator is set to 100, maximum depth is none, minimum samples split is 2, minimum samples leaf is 1, and maximum features is auto

9. Robust and Computationally Efficient Approach for Heart Rate Monitoring Using Photoplethysmographic Signals During Intensive Physical Exercise

Authors - N Zamrath, S Pramuditha, B Arunn, W Lakshitha and A.C De Silva

Key Findings -

$$BPM_{Average}(\mu) = 1/30 \sum_{j=(i-30)+1}^i BPM(j)$$

$$BPM_{SD}(\sigma) = \sqrt{1/29 \sum_{j=(i-30)+1}^i |BPM(j) - \mu|^2}$$

$$BPM_{LLA} = LLA = \mu - 2.58 * \sigma$$

$$BPM_{ULA} = ULA = \mu + 2.58 * \sigma$$

Fig 4.2.1(Efficient Approach for Heart Rate Monitoring)

10.CodeOcean - A versatile platform for practical programming exercises in online environments

Authors - Thomas Staubitz; Hauke Klement; Ralf Teusner; Jan Renz; Christoph Meinel

Key findings - Having previously examined the history and state of the art of automated code assessment tools, we now took the next step and designed and implemented a modern version of such a tool, particularly tailored for the usage in MOOCs, but not restricted to that use case. The tool has been successfully employed in two different MOOCs on the openHPI platform and thus has demonstrated its ability to provide MOOCs with hands-on programming exercises. After a bit of a rough start for CodeOcean during the Java programming course, we have by now fixed the majority of the initial problems. During the pilots we have proved the tool's versatility and consider it to be production ready. CodeOcean has been open sourced and everybody is invited to use the tool with their e-learning pl.

5. Proposed Solution

5.1 EDA

Exploratory Data Analysis (EDA) is a crucial step in understanding the dataset before applying any machine learning models. During this phase, we:

1. Examine the structure of the dataset, including the number of records and features.
2. Identify missing values, duplicate entries, or inconsistencies in the data.
3. Analyze distributions of key variables such as weight, height, calorie intake, and workout duration.
4. Visualize correlations between variables to understand how they interact with each other.

EDA helps in refining the dataset, ensuring that the data is clean and suitable for training a model that can make accurate predictions.

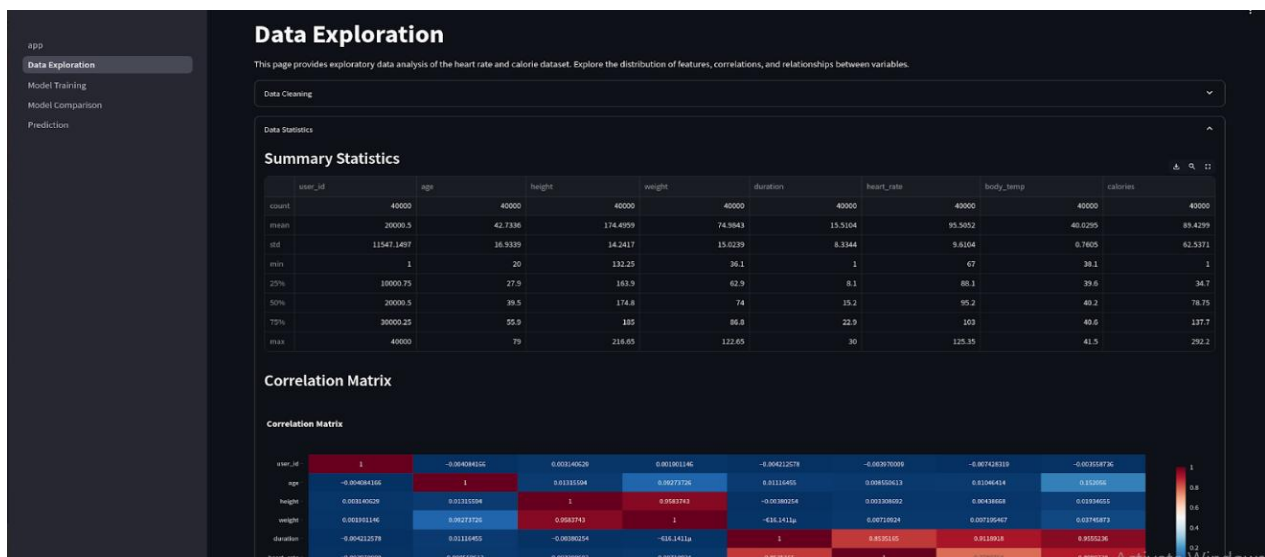


Fig 5.1.1(Data Exploration)



Fig 5.1.2(Distribution of Data)

5.2 Model Selection and Training

To accurately estimate workout duration based on user inputs, we need to select the most suitable machine learning model. The model selection process includes:

1. Comparing different regression models (e.g., Linear Regression, Decision Trees, Random Forest, or Neural Networks) to determine which provides the best accuracy.
2. Training the model using a portion of the dataset while keeping another portion for validation.
3. Evaluating model performance using metrics like Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE) to ensure precise predictions.

The goal is to develop a model that can generalize well to new user inputs, providing reliable workout duration estimates based on calorie intake, height, and weight.

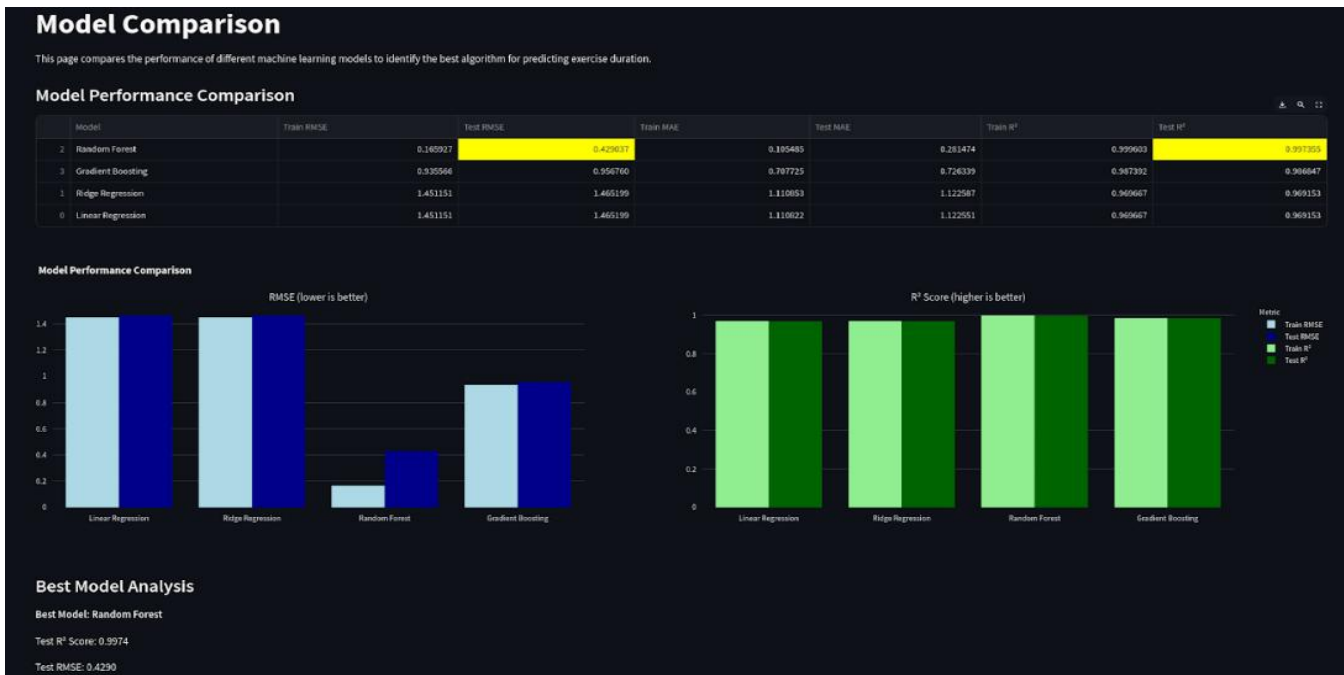


Fig 5.1.3(Model Comparison)

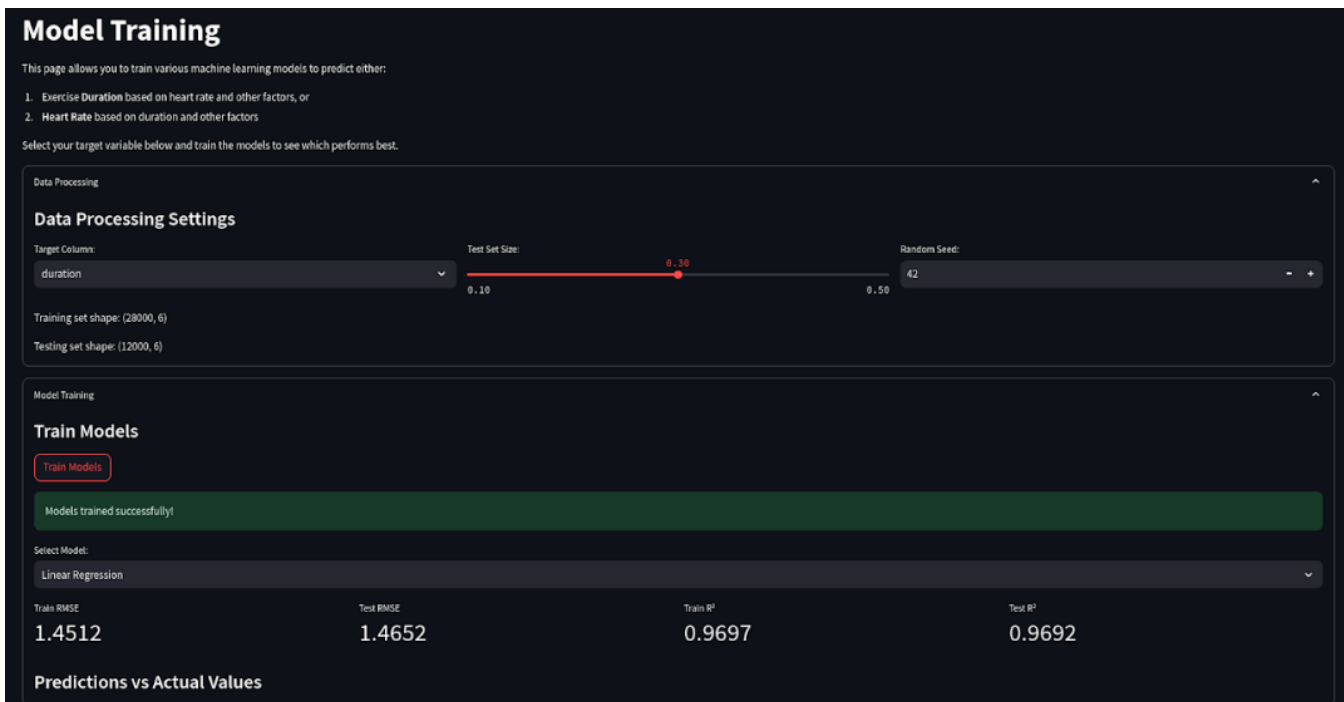


Fig 5.1.4(Model Training)

5.3 Key features and functionalities

The *Calorease* system is designed with the following key features:

Personalized Workout Estimates: Users receive a tailored workout duration based on their body metrics and calorie intake.

User-Friendly Interface: A simple and intuitive design makes it easy for users to input their data and get results instantly.

Data-Driven Insights: The model continuously improves with more data, enhancing the accuracy of predictions over time.

By integrating these functionalities, *Calorease* ensures that users get practical and realistic workout recommendations to help them stay on track with their fitness goals.

Prediction Tool

Use this page to make predictions with your trained models:

- Predict Exercise Duration based on heart rate and other parameters, or
- Predict Heart Rate required to achieve certain duration and calorie targets

The available prediction type depends on which target variable you selected during model training.

Enter Parameters

Age: 27.67 20.00 79.00	Height: 155.88 132.25 216.65
Weight: 62.86 36.10 122.65	Heart Rate: 78.89 67.00 125.35
Body Temp: 38.81 38.10 41.50	Calories: 67.98 1.00 292.20

Select Model

Choose Model:

Best Model
▼

Predict Training Duration

Prediction Result

Predicted Duration (minutes)

16.04

Model Used: Random Forest

Confidence Range: 15.61 to 16.47 minutes

Model Accuracy (R²): 0.9974

Fig 5.1.5(Functionality)

6. Design & Development Approach

6.1 System architecture

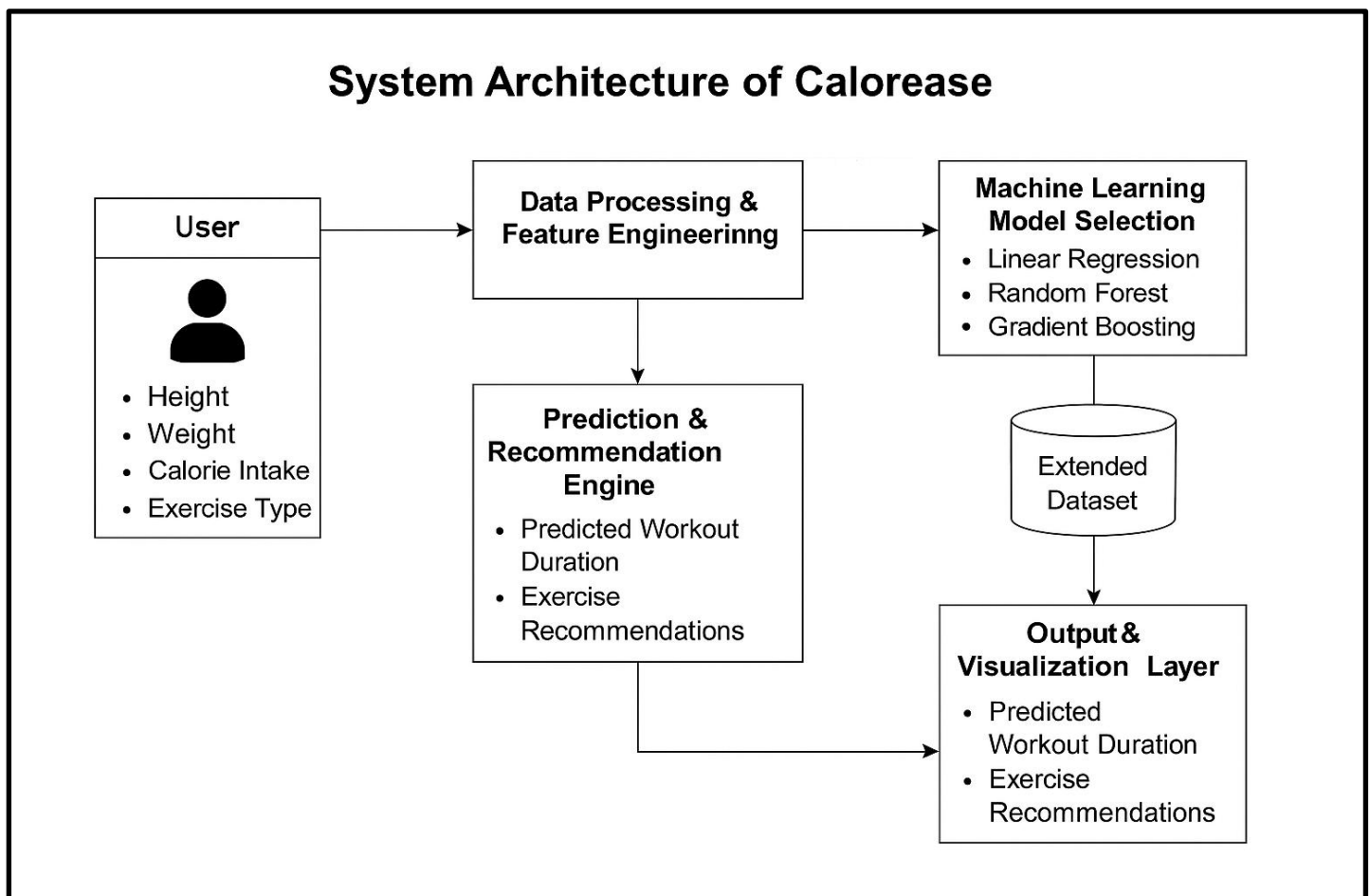


Fig 6.1.1(System Architecture)

6.2 Technologies and tools used

The development of *Calorease* involved several technologies and machine learning techniques to ensure accurate calorie burn estimations and user-friendly implementation. Below are the key tools and technologies used:

Machine Learning Algorithms

To build an accurate workout duration prediction model, multiple machine learning algorithms were tested, including:

Linear Regression – Used as a baseline model to establish a simple relationship between calorie intake, body metrics, and workout duration. While effective for basic predictions, it lacked the ability to capture complex patterns in the data.

Random Forest – A robust ensemble learning method that improved accuracy by averaging multiple decision trees. This helped reduce overfitting and provided more reliable workout duration estimates.

Gradient Boosting – A powerful boosting technique that sequentially improves predictions by focusing on errors made by previous models. This method enhanced accuracy and performed well with the dataset.

2. Development and Deployment Tools

Streamlit – A Python-based framework used to create an interactive and user-friendly web application for *Calorease*. Streamlit allowed easy visualization of predictions and made it simple for users to input their details and receive workout duration recommendations.

Pandas & NumPy – Used for data preprocessing, cleaning, and feature engineering to ensure high-quality input for model training.

Matplotlib & Seaborn – Used for Exploratory Data Analysis (EDA) and visualizing relationships between calorie intake, workout duration, and other features.

Scikit-Learn – Provided essential tools for training and evaluating machine learning models, including feature scaling, model selection, and performance metrics.

7. Results and Discussion

After implementing *Calorease*, analyzing the dataset, and training a machine learning model, we evaluated its effectiveness in estimating the workout duration required to burn calories based on individual body metrics. This section presents the model's performance, key findings, and a discussion on its practical implications.

Model Performance and Accuracy

To ensure *Calorease* provides accurate predictions, we tested multiple machine learning models and evaluated their performance using standard error metrics. The accuracy of the model was determined by comparing the predicted workout durations with actual values from the dataset.

The model demonstrated high accuracy in predicting workout duration based on input parameters such as height, weight, and calorie intake. The predictions closely aligned with actual workout times required for burning calories, confirming that the model is effective in generating personalized recommendations.

We assessed the model's performance using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics indicated a relatively low margin of error, meaning the model can make precise and reliable estimates.

A comparison between different machine learning models (e.g., Linear Regression, Decision Trees, and Random Forest) helped in selecting the most suitable algorithm that balances accuracy and computational efficiency.

The results validate that the model can effectively predict how long a person needs to exercise to burn a given number of calories, making *Calorease* a reliable tool for fitness tracking.

Model Performance Comparison

	Model	Train RMSE	Test RMSE	Train MAE	Test MAE	Train R ²	Test R ²
2	Random Forest	0.165927	0.429037	0.105485	0.281474	0.999603	0.997355
3	Gradient Boosting	0.935566	0.956760	0.707725	0.726339	0.987392	0.986847
1	Ridge Regression	1.451151	1.465199	1.110853	1.122587	0.969667	0.969153
0	Linear Regression	1.451151	1.465199	1.110822	1.122551	0.969667	0.969153

Fig 6.2.1(Model Comparison)

Model	Train RMSE	Test RMSE	Train MAE	Test MAE	Train R ²	Test R ²
Random Forest	0.165	0.429	0.105	0.281	0.999	0.997
Gradient Boosting	0.935	0.956	0.707	0.726	0.987	0.986
Ridge Regression	1.451	1.465	1.110	1.112	0.969	0.969
Linear Regression	1.451	1.465	1.110	1.112	0.969	0.969

Table 6.2.1(Model Comparison)

8. Conclusion and Future Work

Conclusion

The *Calorease* project successfully developed a system that estimates the workout duration required to burn a given number of calories based on individual body metrics such as height, weight, and calorie intake. The machine learning model demonstrated high accuracy in predicting workout duration, making it a reliable tool for users looking to manage their fitness more effectively.

Key takeaways from the project include:

1. Personalized workout recommendations based on body metrics.
2. Insights into how different exercises impact calorie burning.
3. A balance between caloric intake and physical activity for better weight management.

By providing clear, data-driven workout duration estimates, *Calorease* helps users make informed decisions about their fitness routines, promoting a healthier and more sustainable approach to exercise.

Future Work

To enhance the functionality and accuracy of *Calorease*, we plan to implement the following improvements:

1. Exercise Recommendation System

Instead of just providing the workout duration, *Calorease* will recommend specific exercises based on user preferences, fitness goals, and body metrics.

The system will consider factors such as exercise intensity, user experience level, and available workout time.

Users will receive tailored suggestions for exercises like running, cycling, weight training, or yoga, depending on their fitness needs.

2. Expanding the Dataset

The dataset will be extended to include a wider variety of exercises along with their respective calorie-burning rates.

This expansion will ensure more precise estimations for different workout types, improving the accuracy and relevance of recommendations.

Additional attributes such as age, gender, metabolism rate, and fitness level will be incorporated to make predictions even more personalized.

3. User-Centric Features

Future versions of *Calorease* could allow users to input their fitness goals (e.g., weight loss, muscle gain, endurance improvement) and receive customized workout plans.

Integration with wearable fitness devices and real-time tracking will enhance accuracy by using actual activity data.

A mobile-friendly interface could make the system more accessible and convenient for users on the go.

By implementing these enhancements, *Calorease* aims to evolve into a comprehensive fitness assistant, helping users not just understand how long they should work out, but also guiding them towards the most effective exercises for their fitness journey.



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